

Research interests : AI for science tools, generative flow networks, LLM reasoning.

Experience

- Sep 2024 – **Assistant Professor of Machine Learning**, MBZUAI, Abu Dhabi
- 2024 **Senior Researcher**, Technology Innovation Institute, Abu Dhabi
- 2018 – 2023 **Researcher**, Mila, Montréal - Canada
- 2017 **Data Scientist**, Booking.com, Amsterdam

Internships

- 2020 **Researcher**, Google Research, Paris
- 2015 **Researcher**, IBM Research, Singapore
- 2014 **Data Scientist**, Infosys, Bangalore

Education

- 2018 – 2023 **Mila — Université de Montréal**, Montréal - Canada, PhD, Supervisor : Yoshua Bengio
- 2015 – 2016 **École Normale Supérieure**, Cachan - France, M.Sc.
- 2012 – 2016 **École Polytechnique**, Palaiseau - France, B.Sc. + M.Sc.

Selected publications

(* : equal contribution)

- 2026 **FinChain : A Symbolic Benchmark for Verifiable Chain-of-Thought Financial Reasoning**, Zhuohan Xie, Dhruv Sahnan, Debopriyo Banerjee, Georgi Georgiev, Rushil Thareja, Hachem Madmoun, Jinyan Su, Aaryamonvikram Singh, Yuxia Wang, Rui Xing, Fajri Koto, Haonan Li, Ivan Koychev, Tanmoy Chakraborty, **Salem Lahlou**, Veselin Stoyanov, Preslav Nakov. *ACL 2026*
- 2026 **Temporal Contrastive Decoding : A Training-Free Method for Large Audio-Language Models**, Yanda Li, Yuhua Liu, Zirui Song, Yunchao Wei, Martin Takáč, **Salem Lahlou**. *Findings of ACL 2026*
- 2026 **Communication Enables Cooperation in LLM Agents : A Comparison with Curriculum-Based Approaches**, Hachem Madmoun, **Salem Lahlou**. *EACL 2026 (Oral)*
- 2026 **TorchGFN : A PyTorch GFlowNet Library**, Joseph D. Viviano*, Omar G. Younis*, Sanghyeok Choi*, Victor Schmidt, Yoshua Bengio, **Salem Lahlou**. *JMLR 2026(+ NeurIPS 2025 Workshop on Frontiers in Probabilistic Inference)*
- 2025 **Loss-guided Auxiliary Agents for Overcoming Mode Collapse in GFlowNets**, Idriss Malek, Aya Laajil, Abhijith Sharma, Eric Moulines, **Salem Lahlou**. *AAAI 2026*
- 2025 **Improved Exploration in GFlowNets via Enhanced Epistemic Neural Networks**, Sajan Muhammad, **Salem Lahlou**. *ICoIAS 2025*
- 2025 **PORT : Preference Optimization on Reasoning Traces**, **Salem Lahlou***, Abdalgader Abubaker*, Hakim Hacid. *NAACL 2025*
- 2025 **On the Privacy Risks of Spiking Neural Networks : A Membership Inference Analysis**, Junyi Guan*, Abhijith Sharma*, Chong Tian, **Salem Lahlou**. *UAI 2025 (+ oral at ICLR 2025 Workshop ICBINB)*
- 2025 **SVRPBench : A Realistic Benchmark for Stochastic Vehicle Routing Problem**, Ahmed Heakl*, Yahia Salaheldin Shaaban*, Martin Takac, **Salem Lahlou**, Zangir Iklassov. *NeurIPS 2025*
- 2023 **A Theory of Continuous Generative Flow Networks**, **Salem Lahlou**, Tristan Deleu, Pablo Lemos, Dinghuai Zhang, Alexandra Volokhova, Alex Hernández-García, Léna Néhale Ezzine, Yoshua Bengio, Nikolay Malkin. *ICML 2023*
- 2023 **GFlowNet Foundations**, Yoshua Bengio*, **Salem Lahlou***, Tristan Deleu*, Edward J. Hu, Mo Tiwari, Emmanuel Bengio. *JMLR 2023*
- 2023 **GFlowNets and Variational Inference**, Nikolay Malkin*, **Salem Lahlou***, Tristan Deleu*, Xu Ji, Edward Hu, Katie Everett, Dinghuai Zhang, Yoshua Bengio. *ICLR 2023*

- 2023 **DEUP : Direct Epistemic Uncertainty Prediction**, **Salem Lahlou***, Moksh Jain*, Hadi Nekoei, Victor Butoi, Paul Bertin, Jarrod Rector-Brooks, Maksym Korablyov, Yoshua Bengio. *TMLR* 2023
- 2019 **BabyAI : A Platform to Study the Sample Efficiency of Grounded Language Learning**, Maxime Chevalier-Boisvert*, Dzmitry Bahdanau*, **Salem Lahlou**, Lucas Willems, Chitwan Saharia, Thien Huu Nguyen, Yoshua Bengio. *ICLR* 2019
- 2025 **Curriculum-Augmented GFlowNets for mRNA Sequence Generation**, Aya Laajil, Abduragim Shtanchaev, Aya El Mir, Sajjan Muhammad, Eric Moulines, Tiago Silva, **Salem Lahlou**. *NeurIPS 2025 Machine Learning in Structural Biology Workshop*; under review at *UAI* 2026

Service (selected)

- 2026 **Area Chair**, *ICLR* 2026, *ICML* 2026, *NeurIPS* 2026
- 2026 **Local Chair**, *AISTATS* 2026
- 2025–2026 **Program Chair / PC Co-Chair**, *NETYS* 2025; *WebConf* 2026 *PhD Symposium*; *FAISys* 2026 (Xiamen, December)
- 2026 **Main Organizer**, *MBZUAI ML Winter School* 2026
- 2022–2026 **Reviewing**, *ICLR*, *ICML*, *NeurIPS*, *UAI*, *AAAI*, *ECAI*, *TMLR*, *IEEE TPAMI*, *JASA*, *ACL*, *NETYS*, and more
- 2025–2026 **MBZUAI committees**, *Faculty Hiring*, *Admissions (ML + Undergraduate)*, *ML Curriculum redesign (Jan–May 2025)*, *TA Chair (Jan–Oct 2025)*
- 2025–2026 **Jury member**, 2 *PhD Candidacy* juries; 12 *MSc Thesis* juries